



FA report of X-Short issue

JUNGLIM G270QAN01.4

Beckett Tsai

Date 2026.04.07

Problem Statement



● General Information

Brand/Project	JUNGLIM				Defect Phenomenon	X-Short															
ES	Beckett				Defect Rate or Q' ty	1pcs															
AUO Model	G270QAN01.4				Defect Station	FIELD RETURN															
AUO S/N	ZA5A9R78KJZZZS0101				RMA No.	*															
Beol/Module	<table><tr><th>ARRAY_FAB_ID</th><th>FEOL_FAB_ID</th><th>CF_FAB_ID</th><th>BEOL_FAB_ID</th><th>LCM_BEOL_FAB_ID</th><th>JI_FAB_ID</th><th>MODULE_FAB_ID</th></tr><tr><td>L6A</td><td>L6A</td><td>C6A</td><td>L6A</td><td></td><td>M11</td><td>S01</td></tr></table>	ARRAY_FAB_ID	FEOL_FAB_ID	CF_FAB_ID	BEOL_FAB_ID	LCM_BEOL_FAB_ID	JI_FAB_ID	MODULE_FAB_ID	L6A	L6A	C6A	L6A		M11	S01						
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L6A	L6A	C6A	L6A		M11	S01															
Week Code	25/45				Issue Date	2026.03.27															

● Defect Panel S/N Picture



Problem Statement

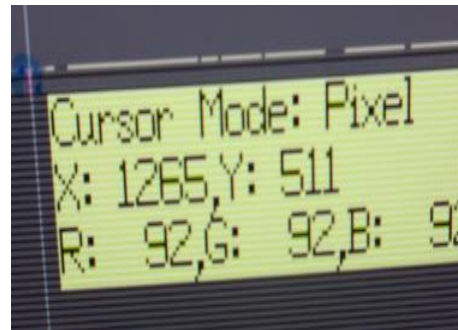


- Describe the phenomenon at customer side

Customer's Feedback Symptom: **H_Block** ◦

- Defect Phenomenon

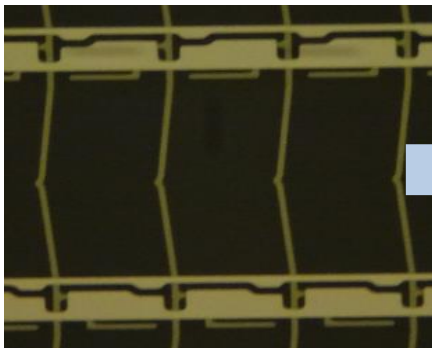
Turned on the panel, and the phenomenon same as customer's feedback and there is bright short_line also found.
ADP internals' defect code: X-Short



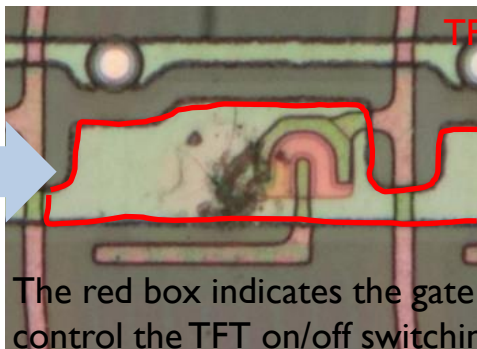
X-Short

Root Cause Analysis

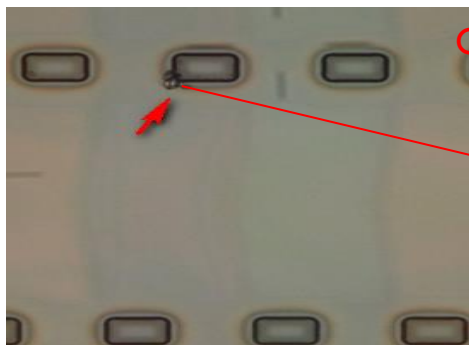
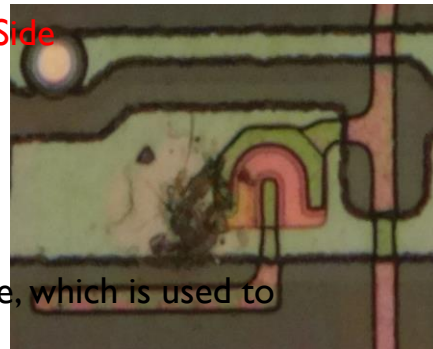
- Step I: After separate the panel, the particle-induced damage was observed on both the TFT and CF sides.
On the TFT side, the gate lines have been severely damaged



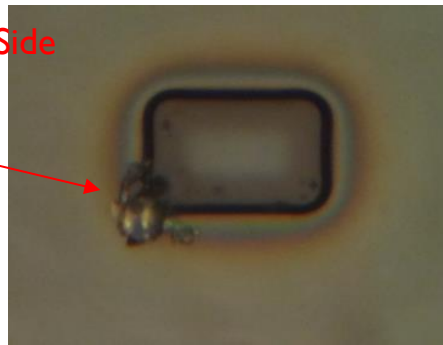
Under Micro-scope



The red box indicates the gate line, which is used to control the TFT on/off switching.

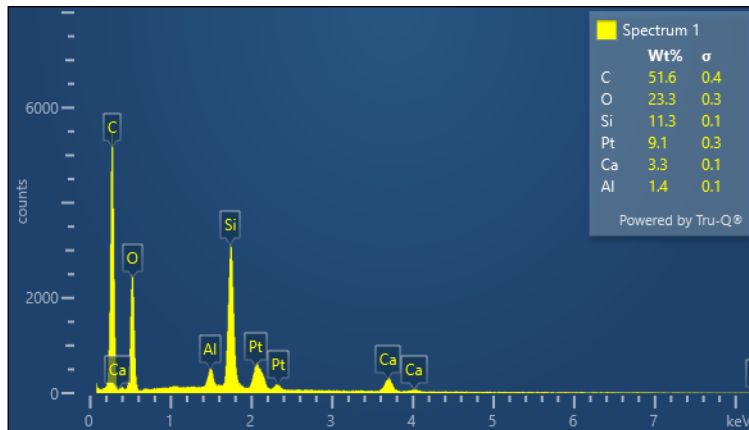
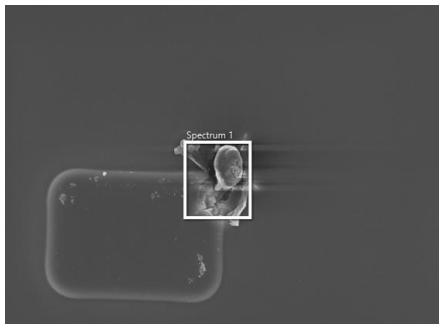


CF Side



Root Cause Analysis

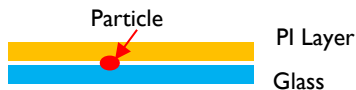
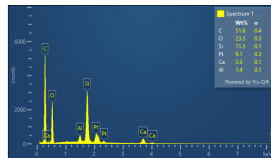
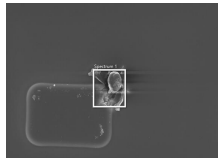
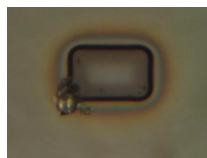
- Step2: EDS showed it contained carbon. (over 50%)



Root Cause Analysis

- Step3: Manufactured History record tracking: 2025/10

ASM_TFT	10/5/2025 12:00:00 AM
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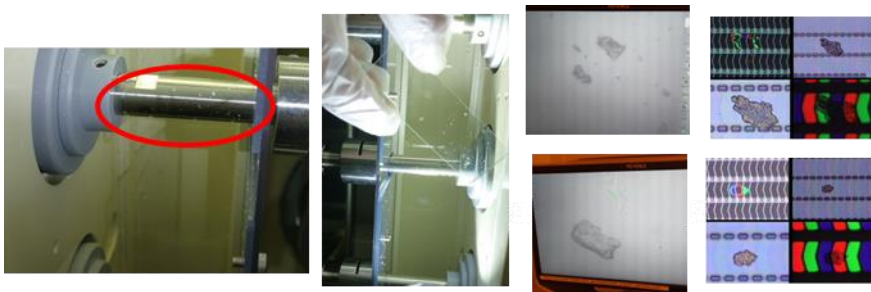
Mechanism Description: Based on the analysis data, the defect is attributed to foreign particle contamination on the upper layer of the PI layer. Elemental analysis indicates that the primary composition is organic material.

Process Analysis: The contamination is located on the upper surface of the PI layer. No rubbing marks or washing flow traces are observed, suggesting that the particle likely fell onto the surface after the alignment cleaning process and prior to cell assembly.

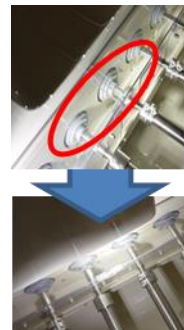
Process Flow: PI printing → PI rubbing → Cleaning → LC + Seal dispensing → TFT + CF Assembly

Countermeasure

- Inspection Findings: Significant wear was observed on the upper surface of the water-stop ring in the cleaning section. Samples were collected for foreign material comparison.
- Foreign Material Characterization: Under OM observation, the particle morphology is similar to the customer-returned defect, exhibiting an irregular, wrinkled, semi-transparent appearance.

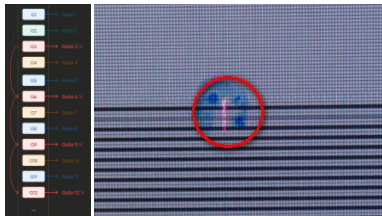


- Replace of water-stop ring components in the cleaning section in 2025/11.
- In_Line Particle Yield loss: 0.4% (2025/10)->0.26%(2025/12)



Summary

- After powering on the returned unit, the phenomenon is consistent with the issue reported by the customer. A short bright line is observed at the upper edge of the defect.
- After disassembling the panel, severe particle-induced scratches were found on the TFT at the location of the bright line (Gate line: used to control TFT on/off).
- As the gate driving scheme of this model is n to $n+3$ (i.e., line 1 to line 4), microscope images confirm that the defect appears at every two-line interval, which is consistent with the design. (refer to the below image)



- Based on the above analysis, the issue is caused by internal foreign particles within the panel, which subsequently affect each corresponding gate line downstream, leading to abnormal behavior.
- The defect is highly suspected to be associated with foreign particles from water-stop ring wear. Improvement has been observed after the countermeasure, and monitoring is ongoing.



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